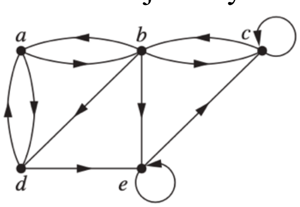
Test3 Review, MAT 2540 Professor Chiu

- This review consists of 7 set of questions.
- You have 60 minutes to complete this review. Show all work and justify your answers.
- Wishing you success.

1. Describe a graph model that represents whether each person at a party knows the name of each other person at the party. Should the edges be directed or undirected? Should multiple edges be allowed? Should loops be allowed?

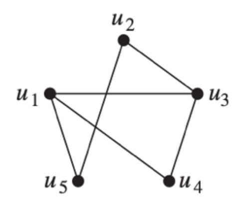
2. Determine whether each of these sequences is graphic. For those that are, draw a graph having the given degree sequence. a) 3, 3, 2, 2, 2, 2 b) 5, 5, 4, 3, 2, 1 c) 5, 3, 3, 3, 3, 3

3. (a) Find the adjacency matrix of the given graph. (b) Find the graph from the given adjacency matrix.

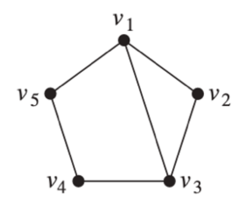


$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

4. Given two graphs G and H. Find the number of vertices, edges, and degrees of vertices respectively and check if they are isomorphic. If yes, then find the isomorphism.



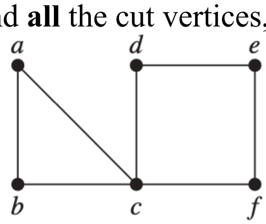
G



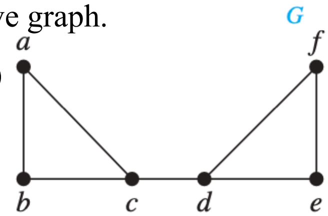
H

5. Find **all** the cut vertices, and **all** the cut edges of each give graph.

(a)

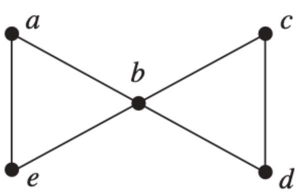


(b)

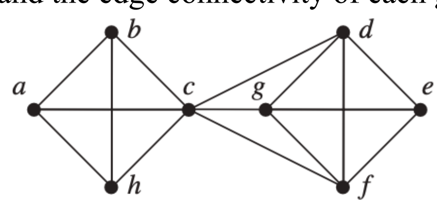


6. Find the vertex cut, the vertex connectivity, the edge cut, and the edge connectivity of each give graph.

(a)



(b)



7. Answer these questions about the rooted tree illustrated.
   (a) Which vertex is the root?
   (b) Which vertices are internal?
   (c) Which vertices are leaves?
   (d) Which vertices are children of j?
   (e) Which vertex is the parent of h?
   (f) Which vertices are siblings of o?
   (g) Which vertices are ancestors of m?
   (h) Which vertices are descendants of b?
   (i) Find the level of each vertex in the given rooted tree. What is the height of this tree?
   (j) Is the given rooted tree a full m-ary tree? Why or why not?